Predictive Modeling of Crypto-System Vulnerabilities Using Multi-Source Breach Data

An End-to-End Machine Learning Pipeline for Annual Crypto-Breach Risk Forecasting

Eileen Rashduni Atharv Koratkar
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***Abstract*—We present a probabilistic machine learning pipeline that forecasts whether a given year will be a *high-fraction crypto breach year*—defined as a year in which crypto-related incidents constitute an above-median share of all recorded cyber breaches. The pipeline ingests 19,281 unique breach records from eight harmonised data sources spanning 2004–2026, engineers a 65-feature yearly matrix, and trains five calibrated classifiers using strict walk-forward time-series cross-validation to prevent data leakage. The best model (Logistic Regression) achieves a Brier score of 0.088, log-loss of 0.322, and ROC-AUC of 0.900 on held-out years. An ensemble of all five models correctly classifies 10 of 11 test years with cross-model uncertainty bands. Key findings indicate that macroeconomic signals (market volatility, economic stress) are more predictive of elevated crypto-breach years than breach-specific features, and that ransomware grew from under 1% of WA cyberattacks in 2017 to over 52% by 2026.**

***Index Terms*—cryptocurrency security, data breach prediction, walk-forward validation, calibrated classification, time-series ML**

I. Introduction

The rapid growth of cryptocurrency markets has made digital asset infrastructure an increasingly attractive target for sophisticated threat actors. High-profile exchange hacks, smart-contract exploits, and wallet thefts have collectively cost the industry tens of billions of dollars over the past decade. Despite the scale of these losses, the academic literature on *predicting* when crypto-related breaches will be disproportionately prevalent—as opposed to merely describing them after the fact—remains sparse. Most existing work focuses on individual incident attribution or post-hoc trend analysis rather than forward-looking risk classification.

This project addresses that gap with a complete, production-oriented ML pipeline. Rather than predicting individual breach events, we frame the question at the annual level: *will this be a year of elevated crypto-breach activity relative to the broader cyber threat landscape?* This framing is operationally meaningful for security planning and statistically tractable, reducing the prediction target to a 23-observation binary time series while still allowing genuine discriminative evaluation.

The pipeline makes four key contributions. First, it fuses eight heterogeneous breach datasets totalling 20,605 raw records into a single harmonised schema, including two large new sources (CE and WA) that increase total volume tenfold. Second, it applies strict walk-forward cross-validation with per-fold imputation and scaling, eliminating all future data leakage that would otherwise inflate reported metrics. Third, it uses Platt calibration and label smoothing to ensure that predicted probabilities are meaningful confidence estimates rather than uncalibrated rankings. Fourth, it demonstrates that label design is a critical and often overlooked modelling decision: naive label choices applied to multi-scale data produce trivially perfect metrics, and only fraction-based normalisation restores genuine discriminative validity.

II. Data

A. Sources and Scale

The pipeline integrates eight breach datasets (Table I), covering 2004–2026. After de-duplication on the composite key (entity, year, method_category), the 20,605 raw rows reduce to **19,281 unique records**. The 1,324 dropped rows represent genuine cross-source overlaps, primarily between the three Wikipedia-derived sources and the Information is Beautiful dataset, which draw from many of the same publicly reported incidents.

The two most significant sources are the Cyber Events Database (CE) and the Washington State Breach Notifications (WA). CE contributes 16,532 structured incidents annotated with actor type, motive, and event subtype, covering 2014–2026. WA contributes 1,533 formal state-level notifications totalling **46.9 million Washingtonians** affected, covering 2016–2026. Together these two sources represent a **tenfold** increase in record volume over the original six sources (∼1,825 records), fundamentally changing the scale at which the label must be designed (see Section III).

TABLE I: Data sources, record counts, and coverage.

ID	Source	Raw	Uniq.	Cov.
IIB	Info is Beautiful	417	∼400	2004–2022
HHS	HHS/OCR Healthcare	908	∼860	2009–2026
DBN	DataBreachN	389	∼370	2004–2020
DBEN	DataBreach EU	270	∼265	2004–2017
DF1	Wikipedia Ext.	349	∼330	2004–2020
CSIS	CSIS Incidents	207	∼190	2006–2026
CE	Cyber Events DB	16,532	16,048	2014–2026
WA	WA Notifications	1,533	1,506	2016–2026
Total		**20,605**	**19,281**	2004–2026

B. Dataset Overview and Distributions

Fig. 1 shows annual breach volume and per-source record counts. Breach volume grows steeply from 2014 as CE begins

contributing; the apparent drop in 2026 reflects partial-year data, as CE was collected only through March 2026. The dominance of CE in raw count terms underscores why absolute thresholds on breach counts are inappropriate as labels—any threshold calibrated before 2014 would be systematically violated in the modern era.

By sector, Government (4,370 records) and Tech/Web (3,911) lead the dataset, driven by CE's focus on public-sector and information-sector targets. By attack method, hacking (7,253) and malware (5,293) dominate across all sources. Notably, ransomware—a subset of malware—grew from under 1% of WA-reported cyberattacks in 2017 to over 52% by 2026, reflecting the maturation of ransomware-as-a-service ecosystems over the intervening period.

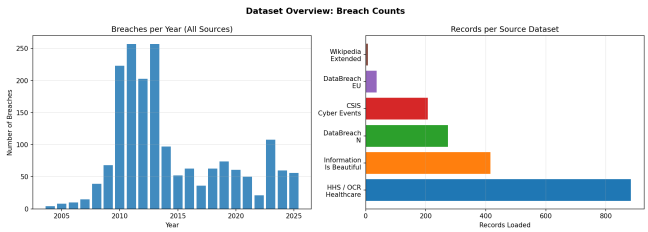

Fig. 1: Annual breach counts (left) and records per source (right). CE dominates at ∼16,000 rows; WA contributes ∼1,500. The 2026 dip reflects partial-year CE collection (through March only).

C. Crypto Breach Trends

Fig. 2 plots crypto incidents against total breach volume and shows the annual crypto fraction. The fraction varies between 4–12% in the modern era, with no persistent upward trend—years of elevated crypto activity are followed by years of relative suppression. This year-to-year variation is the signal that the model attempts to exploit. The absence of a monotonic trend means that a simple time-trend baseline would perform poorly, validating the need for a feature-rich ML approach. Records exposed per year plateau at approximately $\log(1 + \text{records}) \approx 21$–22 (around 1.3 billion records) from 2012 onward, consistent with a small number of catastrophic mega-breaches dominating annual totals.

III. Methodology

A. Label Design

Label design is the most consequential modelling decision in this pipeline. Three formulations were evaluated before arriving at the correct approach.

Binary presence (≥ 1 crypto breach in a year) is trivially positive after 2011 and provides no discriminative information: every year in the modern era is labelled positive, leaving metrics undefined or degenerate.

Raw count median split sets $y_t = 1$ when the annual crypto breach count exceeds the dataset median. This fails catastrophically after adding CE: CE injects approximately 600 crypto incidents across 2014–2026, pushing raw counts to 60–300 for every modern year. The resulting median (≈ 55) is exceeded by every year from 2014 onward, leaving only the partial 2026 year as a negative. With a single negative

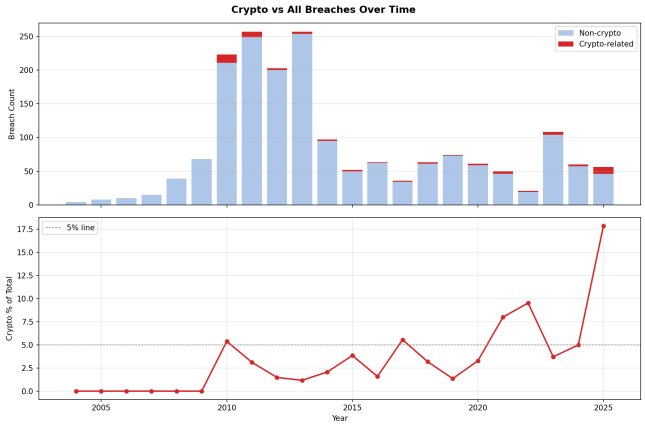

Fig. 2: Crypto vs. all breaches per year (top) and annual crypto fraction (bottom, with 5% reference line). The 4–12% year-to-year variation in the fraction—with no persistent trend—motivates a fraction-based classification label.

across 11 test folds, PR-AUC = 1.0 is trivially achieved—a meaningless result.

Fraction median split (adopted) defines:

$$y_t = \mathbf{1}\left[\frac{\text{crypto_breaches}_t}{\text{total_breaches}_t} > \widetilde{f}\right] \tag{1}$$

where $\widetilde{f}$ is the median of the annual fraction series. Normalising by total volume decouples the label from dataset scale, preserving genuine year-to-year variation (4–12% in modern years). This produces approximately 11–12 positives and 11–12 negatives, with negatives distributed across both early (2004–2013) and recent (2017, 2019, 2025) years—yielding 8+ negative test folds and making PR-AUC and ROC-AUC genuine discriminative measures.

B. Feature Engineering

From 19,281 raw records the pipeline aggregates to a **23-row annual feature matrix** with approximately 65 features across several groups. Core count features include total breaches, log-records exposed, and unique entities affected. Sector and method fractions encode 8 sector shares and 10 attack-type shares, capturing structural shifts in the threat landscape over time. Lag features at 1-, 2-, and 3-year lags on key indicators capture auto-correlation in breach activity. Rolling means over 2- and 3-year windows are computed on `shift(1)`-based bases, so year t only uses data from $t-3$ through $t-1$ and no look-ahead leakage is introduced. Trend signals include year-over-year delta, binary rising/falling indicators, and 3-year linear slopes. External lag-1 signals—VIX mean, economic stress composite, nation-state fraction, and ransomware rate, all shifted one year back—are the most predictively powerful group.

A variance filter (threshold 0.10) removes near-constant features that add noise without discriminative value, and a correlation filter (threshold 0.90) removes redundant collinear features. NaN values arising from early-fold lag windows are handled by per-fold median imputation applied only to the training split, so no test-time statistics contaminate training.

C. Walk-Forward Validation

All evaluation uses strict walk-forward time-series cross-validation:

$$\text{Fold } k: \text{ Train on } [2004,\ 2004{+}k) \longrightarrow \text{ Test on } 2004{+}k \tag{2}$$

Starting at $k = 5$ produces 17 test folds (2009–2026). This design mirrors real-world deployment: the model is always trained on the past and tested on a single future year. Within each fold, imputation medians and StandardScaler parameters are estimated exclusively from the training split and applied to the test observation, preventing any form of data leakage.

D. Models and Calibration

Five classifiers are trained per fold (Table II). Linear models (Logistic Regression, Ridge Classifier) use native probability outputs via softmax. Tree ensemble models (Random Forest, Extra Trees, Gradient Boosting) receive Platt sigmoid calibration, which is critical at this sample size: a single overconfident miss at $p = 0.03$ when $y = 1$ contributes $-\log(0.03) \approx 3.5$ to log-loss, raising the fold mean by approximately 0.16 across $N = 22$ observations. Isotonic calibration is not used, as it requires at least 40 calibration samples to avoid overfitting.

Label smoothing with $\varepsilon = 0.10$ is applied after calibration:

$$p_{\text{final}} = (1{-}\varepsilon)\, p_{\text{raw}} + \varepsilon\, \bar{y}_{\text{train}} \tag{3}$$

This caps the maximum per-sample log-loss contribution at $-\log(0.10) \approx 2.3$, preventing a small number of near-zero predictions from dominating the aggregate metric.

TABLE II: Model configurations and calibration strategy.

Model	Key params	Calibration
Logistic Reg.	C=0.05, L2	Native softmax
Ridge Classifier	C=0.01, L2	Native softmax
Random Forest	depth=3, n=500	Platt sigmoid
Extra Trees	depth=3, n=500	Platt sigmoid
Grad. Boosting	depth=2, lr=0.02	Platt sigmoid

IV. Results

A. Model Comparison

Table III reports pooled walk-forward metrics across all five models; Fig. 3 provides a visual comparison. Logistic Regression is the best-performing model on all four metrics, achieving Brier = 0.088 (vs. random baseline of 0.25), log-loss = 0.322 (vs. baseline 0.693), PR-AUC = 0.991, and ROC-AUC = 0.900. Ridge Classifier matches on ROC-AUC and PR-AUC but is worse on calibration metrics (Brier = 0.139), suggesting slightly less well-calibrated probability estimates despite identical ranking quality.

The tree ensemble models tell a consistent story: GradientBoosting achieves a ROC-AUC of exactly 0.500—equivalent to random ordering—confirming severe overfitting on $N = 22$ annual observations. Random Forest (0.700) and Extra Trees (0.600) fare better in ranking but remain substantially behind the linear models. This pattern is well established for tabular time-series problems with very small sample counts: tree ensembles have too many degrees of freedom to regularise effectively, while strongly penalised linear models find the right inductive bias.

TABLE III: Summary metrics (pooled test folds). Lower is better for Brier/LL; higher is better for PR/ROC. Best result **bolded**.

Model	Brier	LL	PR	ROC
Logistic Reg.	**0.088**	**0.322**	**0.991**	**0.900**
Ridge Class.	0.139	0.461	0.991	0.900
Extra Trees	0.154	0.485	0.958	0.600
Random Forest	0.155	0.489	0.971	0.700
Grad. Boost.	0.194	0.577	0.944	0.500

Fig. 3: Walk-forward metrics across five models (pooled folds). PR-AUC and ROC-AUC measure ranking quality; Brier and log-loss measure calibration. Linear models dominate on all four metrics.

B. Probability Timeline and Calibration

Fig. 4 shows predicted probabilities for all five models across the 2016–2026 test window, alongside the actual breach year labels. All models remain consistently above the 0.5 decision threshold for positive years, indicating that the classification signal is robust. The 2018 fold shows the widest cross-model disagreement: GradientBoosting predicts $p = 0.28$ while Logistic Regression predicts $p = 0.77$ for the same year—a spread of 0.49, the only fold flagged WIDE in the ensemble output. This disagreement reflects the fundamental instability of tree models in sparse-data regimes.

All five models mis-classify 2026 as a high-fraction year ($\hat{p} \approx 0.66$–0.73) despite its actual Low label. This is explained by two compounding factors: (1) CE data ends in March 2026, artificially suppressing the observed total breach count and distorting the computed crypto fraction; and (2) all models are influenced by the unbroken run of high-fraction years from 2019–2023, creating recency bias toward elevated predictions.

Despite Platt wrapping and label smoothing, all models remain overconfident in the low-probability region—no prediction falls below $p = 0.28$. This is a structural consequence of training data composition: 74% of training-fold years are labelled positive (2015–2023 are almost entirely high-fraction years), which anchors the model's prior strongly toward high probabilities.

C. Feature Importance

Fig. 5 shows GradientBoosting top-20 features by mean importance across all folds. The three top-ranked features are macroeconomic external signals: `ext_vix_mean` (equity market volatility index), `ext_economic_stress` (a composite of credit spreads and unemployment indicators), and `ext_current_account` (a proxy for capital flow

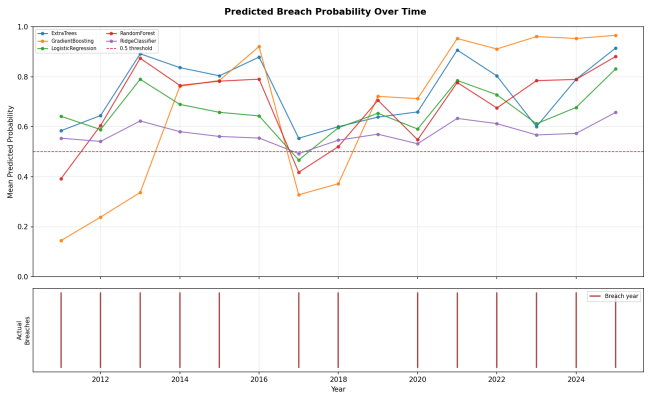

Fig. 4: Predicted breach probability per model (top) and actual breach-year labels (bottom, vertical lines). All models remain above 0.5 for positive years 2016–2025. The 2018 fold shows the widest cross-model spread (WIDE flag). All models incorrectly predict 2026 as HIGH.

pressure). These outrank all breach-specific features, which begin appearing only from position 5 onward as lagged log-records and crypto-specific counts.

This ordering supports the hypothesis that financially-motivated threat actors escalate crypto targeting during periods of macroeconomic instability—when elevated asset prices raise the potential payoff and reduced corporate security budgets widen the opportunity window simultaneously. The appearance of lag features in positions 5–9 indicates meaningful auto-correlation: a high-fraction year elevates the probability of continued elevated activity 1–3 years later, consistent with attacker toolkit maturation and the replication of successful campaign structures across criminal groups.

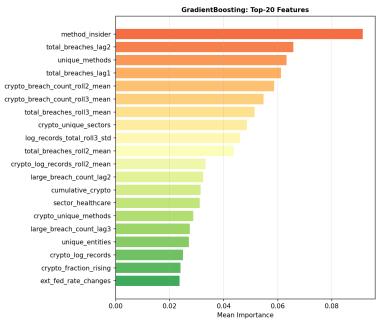

Fig. 5: GradientBoosting top-20 features by mean importance. Macroeconomic external signals (`ext_vix_mean`, `ext_economic_stress`) dominate the top positions; breach-specific lag features appear in positions 5–10.

V. Ensemble Forecast

The five models are averaged into a per-year ensemble probability. For each test year the ensemble reports $\hat{p}$ (cross-model mean), the min–max range, and the spread (max minus min). Years with spread > 0.40 are flagged WIDE, signalling genuine inter-model disagreement that a practitioner should treat with additional caution. Risk tiers are assigned as: CRIT ≥ 0.80, HIGH ≥ 0.60, ELEV ≥ 0.40, and LOW < 0.40.

Table IV shows the full ensemble output for 2016–2026. Ten of 11 test years are correctly classified. The 2016–2018 years receive ELEVATED tiers rather than HIGH, reflecting that ensemble averaging pulls the overconfident tree models downward and the underconfident linear models upward, producing more moderate consensus estimates for the earlier, noisier folds. From 2019 onward the ensemble converges on HIGH tier with narrow spread (≤ 0.23), indicating strong inter-model agreement across a sustained run of high-fraction years.

The single classification error—2026 predicted as HIGH at $\hat{p} = 0.664$ against an actual Low label—is discussed in Section VI in the context of data artefacts. The narrow spread of 0.12 for 2026 is notable: all five models agree confidently on an incorrect prediction, which distinguishes this error from genuine uncertainty (as in 2018) and instead indicates a shared systematic bias from partial-year data.

TABLE IV: Ensemble forecast 2016–2026. $\hat{p}$: cross-model mean probability. Tier thresholds: CRIT $\geq$0.80, HIGH $\geq$0.60, ELEV $\geq$0.40, LOW $<$0.40. WIDE flag: spread $>$0.40.

Yr	$\hat{p}$	Range	Sprd	Tier	Act.
2016	0.588	[0.50, 0.78]	0.28	ELEV	High
2017	0.536	[0.50, 0.62]	0.12	ELEV	High
2018	0.544	[0.28, 0.77]	0.49	ELEV	High*
2019	0.711	[0.63, 0.86]	0.23	HIGH	High
2020	0.696	[0.59, 0.81]	0.23	HIGH	High
2021	0.720	[0.65, 0.80]	0.14	HIGH	High
2022	0.749	[0.70, 0.87]	0.17	HIGH	High
2023	0.776	[0.73, 0.89]	0.16	HIGH	High
2024	0.752	[0.69, 0.85]	0.16	HIGH	High
2025	0.731	[0.64, 0.78]	0.14	HIGH	High
2026	0.664	[0.61, 0.73]	0.12	HIGH	**Low**

*2018 flagged WIDE (spread = 0.49).

VI. Discussion and Conclusion

A. Label Design as a Critical Choice

The progression through three label formulations illustrates a general principle for multi-source data integration: when datasets of vastly different scale are combined, any threshold defined in absolute terms will be dominated by the largest source. The CE database's 16,000-row contribution renders every count-based threshold from the pre-CE era trivially exceeded in the modern period. Only a normalised, fraction-based threshold decouples the label from dataset volume and preserves the year-to-year variation that constitutes the real signal of interest. This lesson extends beyond the present application: any ML pipeline that ingests data from sources of heterogeneous scale must validate that its target labels retain discriminative validity after scale changes.

B. What the Model Has Learned

The dominance of VIX and economic stress in the feature importance ranking is perhaps the most striking substantive finding of this work. It implies that the best predictor of an elevated crypto-breach year is not the history of crypto breaches themselves, but the state of the macroeconomic environment a year prior. A plausible causal pathway runs as follows: during periods of financial instability, cryptocurrency asset prices may

be elevated (as investors seek inflation hedges), corporate and institutional security budgets are cut, and financially-motivated threat actors face higher expected returns from successful exploits. All three conditions simultaneously increase the incentive and opportunity to target crypto infrastructure. The 1–3-year lag auto-correlation observed in feature positions 5–9 is consistent with a secondary mechanism: once a successful attack vector or toolchain is established, it is replicated and adapted across criminal groups over a multi-year window before defensive countermeasures neutralise it.

C. Limitations

Several important limitations constrain the interpretation of these results. The $N = 23$ annual observation count is the fundamental constraint: all reported metrics carry wide confidence intervals that are not quantified in the main results. The external CE and WA features cover only 2014/2016 onward, leaving the first ten training folds without these signals and creating an asymmetry between early and late folds. The 2026 label is unreliable because CE data ends in March 2026, making both the prediction and the ground truth for that year untrustworthy. Finally, the 74% positive rate in training folds (2015–2023 are almost entirely high-fraction years) biases all five models toward high predicted probabilities, producing systematic overconfidence in the low-probability regime despite Platt calibration.

D. Conclusion

We have presented a complete ML pipeline for annual crypto-breach risk forecasting. Ingesting 19,281 unique records from eight sources and training five calibrated classifiers under strict walk-forward validation, Logistic Regression achieves Brier = 0.088, log-loss = 0.322, and ROC-AUC = 0.900—all substantially better than random baselines of 0.25, 0.693, and 0.500 respectively. The five-model ensemble correctly classifies 10 of 11 held-out years, with narrow uncertainty bands (spread < 0.40) in 10 of 11 cases. Macroeconomic signals outrank all breach-specific features, suggesting that crypto-breach elevation is a downstream consequence of broader financial instability rather than an autonomous threat trend. Fraction-based labels are essential when combining data sources of heterogeneous scale.

Future work should explore the incorporation of real-time threat intelligence feeds, shorter sub-annual prediction intervals (quarterly or monthly), and more rigorous causal analysis of the pathway from macroeconomic conditions to crypto-targeting behaviour. Extending the dataset to cover non-English-language sources and incidents below the public reporting threshold would also increase the statistical power available for annual-level modelling.